Anomaly Detection with Python – Summary

Master's in Data Science – Machine Learning project. Code: `src/utils.py`; analysis: `notebooks/anomaly_detection.ipynb`.

1. Protocol

- **Organisation**: all the logic (preparation, training, scoring, threshold selection, evaluation, plots) lives in `utils.py`; the notebook only contains short calls and the analysis. Seed 42, library versions printed at the top (Python 3.12.3, scikit-learn 1.9.1, imbalanced-learn 0.14.2, shap 0.54.1), training and scoring times measured, warnings not silenced.
- **Credit Card**: 1,081 exact duplicates (19 of them frauds) removed *before* splitting so that no transaction can appear in two partitions (283,726 rows, 473 frauds). `Time` excluded from the features. Stratified 60/20/20 split: 170,235 / 56,745 / 56,746 transactions (284 / 94 / 95 frauds), identical for every method.
- **Preprocessing fitted on the training set only** (inside a `Pipeline`): `log1p(Amount)` + `StandardScaler` for LOF, logistic regression and SMOTE (distance- or gradient-based methods); nothing for the tree and Isolation Forest (scale-invariant). SMOTE (ratio 0.1) applied to the training set only; validation and test never rebalanced.
- **Harmonised scores** (higher = more suspicious: sign of `score_samples` flipped for IF and LOF). **Ranking**: AP (main metric) and ROC-AUC. **Decision**: threshold chosen on **validation** = maximum recall subject to precision ≥ 80 % (fraud) or ≥ 90 % (maintenance), at least one alert; otherwise "target not met" and best trade-off (max F1). Model and threshold are frozen before the test set, which is used only once.
- **≤ 6 configurations per family**; LOF (expensive) fitted on a fixed stratified subsample of 30,000 rows (50 frauds). IF and LOF are fitted without labels; choosing their hyperparameters and threshold with validation labels is still a supervised selection step.

2. Main results

Mouse (500 points, 10 anomalies). Near-perfect ranking: AP 0.961 (IF) and 1.000 (LOF). IF's default threshold flags 125 points (F1 0.15); the IQR rule (k = 1.5 and 3, fixed a priori) improves its F1 (0.65 and 0.75) but degrades LOF's, already perfect by default (1.00 → 0.43 / 0.71). Changing the threshold changes neither AP nor AUC, which only depend on the order of the scores. In novelty detection (LOF `novelty=True` fitted on 392 normal points, test set of 98 normal + 10 anomalies): AP 1.0, F1 0.83 (4 false alarms). These results rest on only 10 anomalies.

Credit Card – test set (thresholds frozen on validation)	AP	AUC	Prec.	Recall	Alerts	FP	FN	≥ 80 %
★ **Weighted logistic regression, C = 1** (selected on validation)	0.726	0.962	**0.882**	**0.789**	85	10	20	yes
Decision tree, SMOTE, depth 5	0.650	0.895	0.804	0.779	92	18	21	yes (barely)
EasyEnsemble (10 subsamples, AdaBoost 50) – Part 3	0.717	0.975	0.768	0.768	95	22	22	no
LOF, k = 100 (30,000-row subsample)	0.403	0.956	0.463	0.663	136	73	32	no
Isolation Forest, 100 trees, max_samples = 4096	0.237	0.955	0.201	0.442	209	167	53	no

- **Validation**: all logistic regressions and 3 trees out of 6 meet the target; class weighting saturates the tree's scores (target not met). The weighted logistic regression is retained (recall 0.766, precision exactly 0.800, 90 alerts). **No unsupervised detector reaches 80 %** (max precision 0.43 for LOF at its trade-off).
- **In transactions**: out of 56,746 transactions, the team reviews **85 alerts** (1.13 alerts per fraud found) and **20 frauds out of 95 go undetected**. The constraint is met on the test set (0.882), but the 95 % Wilson interval of precision, [0.80, 0.94], touches the limit.
- ROC-AUC (0.94 to 0.98 for every method) does not separate the methods; AP and the precision-recall curve do. The maxima of F1 and especially balanced accuracy (1,000 to 1,900 alerts) do not match the precision constraint.

3. Researched method: EasyEnsemble

Source: X.-Y. Liu, J. Wu, Z.-H. Zhou, "Exploratory Undersampling for Class-Imbalance Learning", *IEEE Trans. SMC-B*, 39(2), 2009; docs `imblearn.ensemble.EasyEnsembleClassifier`. **Principle**: *T* balanced subsamples (all frauds + as many random normal rows), one AdaBoost per subsample, score = mean probability. Limits the information loss of a single undersampling, without synthetic data. Hyperparameters studied: *T* and the number of AdaBoost stumps (3 configurations). **Success**: the only Part 3 method to meet the constraint on validation (precision 0.800, recall 0.723); AP 2 to 5 times that of LOF and IF. **Limit**: supervised (labels required); threshold chosen at the edge of the constraint, hence a test precision of 0.768 (constraint not met, 95 % CI [0.67, 0.84]); no better than logistic regression. **Cost**: ≈ 2 s of training. As a complement (validation only, no extra dependency): ECOD (AP 0.249) and a Gaussian mixture (0.188) stay below LOF (0.337).

4. Transfer to predictive maintenance (AI4I 2020) and robustness

- **AI4I** (10,000 synthetic observations, 339 failures = 3.4 %): identifiers and TWF/HDF/PWF/OSF/RNF dropped (target leakage); Type one-hot encoded, numeric features standardised (fitted on train); same hyperparameters as Credit Card, models refitted, thresholds re-chosen. Validation: logistic regression precision 0.917 with only **12 alerts** (recall 0.16). **Test**: 4 interventions, all useful, 64 failures missed out of 68 (constraint met, but CI [0.51, 1]); IF: 3 interventions including 1 unnecessary (0.667, not met).
- **Why the winning method transfers poorly**: heterogeneous failures driven by non-linear interactions (torque × speed, temperature difference) that a linear model captures poorly; 20 times higher prevalence. APs are not

comparable across datasets: relative to prevalence, 434 × random for fraud vs 12 × for maintenance.
- **Robustness (Mouse, 3 seeds, real test set unchanged)**: LOF goes from 1.00 to 0.986 (uniform generator) or 0.964 (local) with 10 % contamination; IF goes from 0.756 to 0.936 or 0.852. The conclusion depends on the generator: synthetic anomalies resemble the real ones and favour IF.
- **SHAP on IF**: `TreeExplainer` explains the *mean path length* (verified by additivity), whose sign is opposite to the score; signs checked with `KernelExplainer`. Detected frauds: very low V14, very high V11; false alarms: high `Amount`, V5, V27; missed frauds: weaker signature or ordinary transaction. A contribution is neither a cause nor a business meaning (anonymised features).

5. Constraints met and limits of generalisation

- **Fraud (≥ 80 %)**: met on validation and on test by the selected method; not met by IF, LOF or EasyEnsemble on test. **Maintenance (≥ 90 %)**: met, but on 4 test alerts only: a very fragile result with 6 % recall.
- **Limitations**: random split although the data is temporal (48 h): results probably optimistic, a time-based split would be needed for prospective use; few positives (94-95 frauds, 68 failures; one fraud = 1.1 % of recall); a single split and a single seed; thresholds chosen at the edge of the constraint (optimistic validation precision); synthetic AI4I data; 2-D Mouse with 10 anomalies.